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CPT202

Ethical and Professional issues

Ethics

- The word: `` a set of **moral** principles : a theory or system of moral values’’
About good and bad; right and wrong
In theory: moral philosophy
Applications: war, environment, animal rights, research

This lecture focuses on:

- Academic ethics
What are researchers, academic members and students expected to practice
- Computer Ethics
Is an IT system of moral standards or moral values used for computer users ?

Academic Ethics Code of Conduct

- Plagiarism, Cheating, and Fabrication of Data.

XJTLU e-Bridge - >

Student Academic Services/ Quick reference ->

Policies and Regulations ->

- ✓ Academic Integrity Policy
 - ✓ POLICY ON STUDENT CONDUCT AND DISCIPLINE
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- Offences can result in attendance at a University-level committee and penalties being imposed.
 - Prevention: Turnitin, citation, quotation ""



Ethical conduct in Research

- XJTLU ‘Policy on Ethical Conduct in Research’ by *University Ethics Committee*
 - OPENNESS AND HONESTY
 - INFORMED CONSENT
 - RIGHT TO WITHDRAW
 - PROTECTION FROM HARM
 - CONFIDENTIALITY
 - DISSEMINATION OF FINDINGS
 - USE OF DATA
- Case study

A research project in “e-learning for kids”:
Need an ethical approval? Why or why not

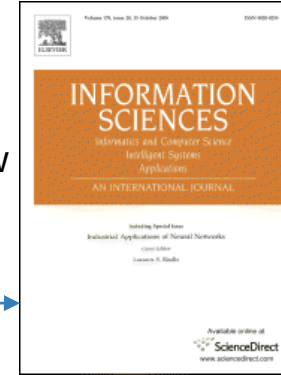


Ethical conduct in Research

Case study – academic paper submission



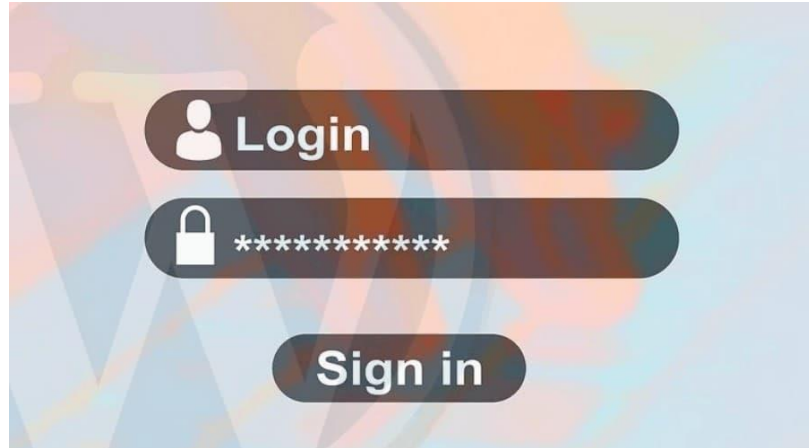
Manuscript submitted for review



Computer Ethics

- Code of Ethics: a guideline in ICT that determines whether a specific computer action is ethical or unethical
- Computer Crime: any illegal acts involving computers
- Cyber Law: to protect the Internet and other online systems
- Case studies - Hack others' Bank Accounts

The common approaches and prevention



Intellectual Property

- Legal protections:

copyrights, trademarks, trade secrets, and patents

- Categories:

Inventions, trademarks, logos, and industrial designs

Copyrighted material (Literary and artistic works, online materials)

- Fair Use Doctrine: for certain purposes

- IT impact on IP:

Easily download / distribute video, not asking for permission

Easily modify others work

Easily copy and paste

How to analyze ethical issues

- What are the facts

(Publishers sued Xerox corporation because users copied books and magazines. The publishers lost \$).

- What are the conflict interests or dilemma
- Who are involved: the stakeholders
- What are the options
- What are the consequences

Ethical Principles

- **Golden Rule:** “Do unto others as you would have them do unto you” (Luke 6:31)
in Chinese “己所不欲，勿施于人” 《论语》
- **Immanuel Kant’s Categorical Imperative:** “If an action is not right for *everyone* to take, then it is not right for *anyone*”
- **Descartes’ rule of change:** “If an action cannot be taken *repeatedly*, then it is not right to be taken at *any time*”

Ethical Principles (Continued)

- **Utilitarian Principle:** “Take the action that achieves the *greatest* value for *all* concerned”
- **Risk Aversion Principle:** “Take the action that produces the least harm or incurs the *least cost* to all concerned”
- **Ethical no-free-lunch rule:** “Assume that all tangible and intangible objects are owned by someone else, unless shown the contrary.” “If someone has created something of value to you, that person probably wants compensation for your use”

Professional issues in IT

- What is professionalism?

Merriam-webster `` the conduct, aims, or qualities that characterize or mark a profession or a professional person

- Professional issues in IT, including
 - Professional Communication: Research, Writing and Presentation
 - Professional Ethics
 - Legal, Social and Cultural Issues
 - IT Governance: principles, Change and Risk Management
 - Teamwork Concepts and Societal Issues

Ethics Codes for IT Professionals

- ACM Codes of Conduct
- IEEE Code of Ethics

Case study

Has the e-calculator destroyed people's ability to do mental arithmetic?

Does this matter? (cf: “Nine Tomorrows” - Asimov)



Case study

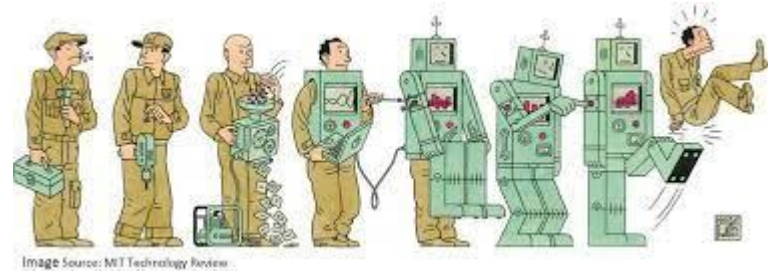
Will intelligent robots devastate to the human.

Is it ethical to do research in AI?

(c.f. Sun Microsystems cofounder Bill Joy's

paper “Why the future doesn't need us”)

Taken from “A Gift of Fire” by Sara Baase



References

- <https://www.scu.edu/ethics/ethics-resources/ethical-decision-making/what-is-ethics/>
- <https://iep.utm.edu/ethics/>
- <https://techviral.net/methods-hackers-use-to-hack-bank-accounts/>
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- Any image taken from internet open domain is for education purposes

Thank you for your attention !

In this lecture, you have learned:

- Ethical issues
- Professional issues
- Cases studies